Armaan Sandhu

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Education

Worcester Polytechnic Institute | *Worcester, MA* **2025 – Present**
B.S. in Computer Science, M.S. in Artificial Intelligence | GPA: 4.0 | Expected May 2029
Carmel Junior College | *Jamshedpur, India* | *Valedictorian* **2010 – 2025**

Technical Skills

Languages: Python, C, C++, Rust, Zig, Java, x86-64 Assembly, JavaScript
Tools: Git, GDB, Linux, Bash, Valgrind, llama.cpp, Redis, PostgreSQL, SQLite
Concepts: OS Kernel Development, Systems & Concurrent Programming, Binary Analysis
AI/LLM: MCP, LLM Routing & Gateways, RAG, Vector Search, Tokenizers, Spec Driven Development

Open Source Contributions

Systems & Security (systemd, util-linux, sudo, curl) **May 2026 – Present**

- 22 merged systemd PRs fixing memory-safety and parsing bugs. Made journalctl reads 1.7–2.5× faster and `-verify` 2.65× faster. Sped up `list-dependencies -all` by over 100×.
- 5 merged util-linux PRs, including a hash-indexed libmount path cache. Merged fixes in the Linux kernel, sudo, curl, CUPS, unbound, and other network and security projects.
- Credited upstream for a sudoedit wildcard policy bypass report and a strongSwan fix.
- Responsibly disclosed several additional vulnerabilities in widely deployed software to maintainers, with fixes in progress.

Browser Engines & AI Infra (Lightpanda, LiteLLM, LanceDB, Lance, CPython) **May 2026 – Present**

- Merged 7 LiteLLM, 9 LanceDB and Lance, 3 Lightpanda, and 1 CPython PRs. Fixed production data-loss and request-cancellation bugs and added universal gateway support for several AI agent harnesses.

70+ pull requests merged across 15+ open-source repositories, spanning C, Rust, Python, and Zig.

Experience

Founding Engineer & Product Head | *MLRouter* **June 2026 – July 2026**

- Built an OpenAI-compatible gateway for 400+ models with cost/latency routing, provider failover, and enterprise governance. Built TokenDrift, a model-upgrade check that measures shifts in token counts, costs, and vocabulary before changes ship.

Open Source Author & Maintainer | *mcp-persist* **May 2026 – Present**

- Built and maintain a production-grade MCP persistence layer across Redis, SQLite, and PostgreSQL, reaching **10k PyPI downloads in its first month** and 600+ tests.

CS Peer Learning Assistant, Peer Mentor & Community Advisor | *WPI* **August 2026 – Present**

- Support undergraduate students through foundational computer science coursework, including data structures and systems programming, while providing academic and personal guidance to residential students.

Research Experience

Independent Study, RunixOS | *WPI, advised by Prof. Craig Wills* **Early 2026 – July 2026**

- Built a capability-based OS kernel from scratch, implemented a preemptive scheduler, synchronous/asynchronous IPC, and a capability-gated filesystem to study OS design under capability security.

Projects

agentfence (Rust, C) and patchproof (Python) **October 2026**

- agentfence: least-privilege sandboxing for AI coding agents that traces a run with strace, never grants secrets, and enforces the learned permissions with unprivileged Landlock.
- patchproof: returns evidence-based verdicts (PROVEN, NO_OP, CANNOT_REPRODUCE, and others) on whether a fix or review suggestion is needed.

codebase-rag and rag-timetravel (Python) **June 2026 – Present**

- Built a codebase intelligence tool for AI coding agents that answers cross-file questions in one query and keeps its index synchronized with code changes. Its debugging engine, rag-timetravel, rewinds searches to the exact data version used and replays queries to show why results changed.

Binary Exploitation & Cache Simulator (C, x86-64 ASM) | *WPI, Machine Organization & Assembly*

- Reverse-engineered a multi-stage binary and built an N-way set-associative LRU cache simulator to validate matrix-transpose optimizations.